MIL-P-45685A(MU)
12 February 1968
SUPERSEDING
MIL-P-45685(Ord)
25 October 1960

MILITARY SPECIFICATION

PERISCOPE: M23

1. SCOPE

1.1 This specification covers one type of periscope known as Periscope: M23.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

MMM-A-131	Adhesives, Glass to Metal, for Bonding of Optical Elements

Military

MIL-G-174	Glass, Optical
MIL-C-675	Coating of Glass Optical Elements
MIL-M-13508	Mirrors, Glass, Front Surface Aluminized for Optical Elements
MIL-O-13830	Optical Components of Fire Control Instruments: General Specification Governing the Manufacture and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel: Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-B-60621	Bonding Optical Assemblies (Glass to Metal)

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

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DRAWINGS

U.S. Army, Frankford Arsenal

F7560085	Fixture, Vibration
D7560124	Milli-Dioptometer 20X
F7633306	Periscope: M23
F7680617	Fixture, Shock Test
F7680673	Fixture, Pressure Test
F7695236	Vibration, Adapter
F7695431	Fixture, Shock Test Adapter
F7695464	Fixture, Final Inspection
F7695490	Fixture, Pressure Test Adapter
C7695507	Fixture, Pressure Test Adapter
F7695508	Fixture, To Measure Power and Definition

PACKAGING DATA SHEET

7633306	Packaging of Periscope: M23

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The periscope shall be manufactured in accordance with Drawing F7633306 Periscope: M23 and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to and compliance with the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The periscope shall show no evidence of physical failure and shall meet all requirements of this specification

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at plus 60 degrees to 90 degrees Fahrenheit after having been exposed and thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.3.2 Vibration.- The periscope shall be vibrated at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration the periscope shall show no evidence of loose or damaged parts and shall meet the requirements of 3.4, 3.5 and 3.6.

3.4 Cleanliness.- The periscope shall meet the cleanliness requirements of MIL-O-13830. The interior optical surfaces of the periscope shall show no more than six particles of dirt. The area of each particle shall not be in excess of the area equivalent to a size fifty dig. There shall be no other dirt within the periscope which shall adversely affect observation and image quality.

3.5 Sealing.- The periscope shall show no evidence of leakage when subjected to an external pressure of five pounds per square inch gage pressure (psig) for a minimum period of five minutes.

3.6 Performance.

3.6.1 Collimation.- With the mounting surface of the periscope in a vertical plane and with the line joining the center of the pilot diameter (4.498-0.003) to the upper guide hole oriented in a vertical direction (see Drawing D7633307, Sheet 2), an incident horizontal line of sight shall be depressed 2° 15' ± 0° 2' and be perpendicular to the mounting surface within 0° 2' in azimuth, when emerging from the exit window.

3.6.2 Spherical power and astigmatism.- The spherical power or astigmatism of the periscope shall not exceed five millidiopters when using a telescope of at least one inch aperture.

3.6.3 Image tilt.- The image of a plumb line shall be vertical within one degree.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-O-13830 and MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

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4.2 First article (initial production) submission.- When required by the contract, an initial production sample shall be submitted to a Government approved facility in accordance with the contract schedule for test and evaluation. The sample shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions herein using approved methods, procedures and equipment to determine conformance to component drawing requirements. In the absence of specific test methods and procedures, the methods and procedures in MIL-O-13830 and MIL-F-13926 shall be utilized as applicable. Certified test results, identifiable with the examination, may be accepted in lieu of repeated tests if such tests will be distructive to the end item or its components. The Government reserves the right to require retests when deemed necessary to assure acceptable components.

4.3.2.1.1 Certification provisions.- Certification shall include test data and results for the characteristics for which certification is acceptable (see 4.4.3). Certification shall be provided prior to performing final acceptance inspection and will suffice for the remainder of the contract, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed. Any and all changes will require a new certification from the contractor.

4.3.2.2 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I and II herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I and II shall constitute the minimum inspection to be performed by the supplier after first article approval, compliance with the certification provisions of this specification and prior to Government acceptance or rejection by item or lot.

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TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC	REQ'T.	TEST PROCEDURE
CRITICAL:	NONE.		
MAJOR:	AQL 0.65 percent defective		
101.	Vibration	3.3.2	4.6.2, 4.6.2.1
102.	Cleanliness	3.4	4.6.3
103.	Collimation	3.6.1	4.6.5.1
104.	Image tilt	3.6.3	4.6.5.2
105.	Spherical power and astigmatism	3.6.2	4.6.5.3
MINOR:	NONE.		

NOTE: Tests in Table I shall be at the standard ambient temperature (60° to 90°F).

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Control tests.

4.4.1 Special sampling.- Three periscopes shall be selected at random as a sample from each 50 produced, or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II and shall then meet the requirements and tests in Table I.

TABLE II

CHARACTERISTIC	REQ'T	TEST PROCEDURE
301. Storage temperatures	3.3.1	4.6.1
302. Sealing	3.5	4.6.4

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.4.3 General.- Unless otherwise specified, certification shall be provided for the characteristics in Table III. Certification shall be in accordance with the provisions in 4.3.2.1 and 4.3.2.1.1.

TABLE III

CHARACTERISTIC	REQ'T	TEST DATA TO COMPLY WITH
303. Fabrication (general)	3.1	Applicable drawings MIL-F-13926 and MIL-O-13830.

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	Fabrication (components)	3.1	Applicable drawings and specification:
304.	Optical glass		MIL-G-174
305.	Optical element bonding		MIL-B-60621
306.	Optical element surface quality		MIL-O-13830
307.	Optical element surface coating		MIL-O-13830
308.	Optical element aluminized surface		MIL-M-13508
309.	Optical element polished surface		MIL-O-13830
310.	Optical element adhesive		MMM-A-131
311.	Optical element performance		MIL-O-13830 and 4.6
312.	General specification	3.2	Applicable drawings and MIL-F-13926

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the List of Inspection Documents (IEL) when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.- Special test equipment in Table IV, comparable to referenced drawings and having at least the accuracy depicted thereon, shall be provided. It shall be utilized to conduct periscope and component tests related to the examinations listed herein.

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TABLE IV

TEST EQUIPMENT	DRAWING	EXAMINATION
Fixture, Final Inspection	F7695464	Periscope (only)
Fixture, To Measure Power	F7695508	Periscope and components
Milli-Dioptometer 20X	D7560124	Periscope and components
Fixture, Vibration Test	F7560085	Periscope (only)
Adapter, Vibration Test	F7695236	Periscope (only)
Fixture, Pressure Test	F7680673	Periscope (only)
Adapter, Pressure Test	F7695490	Periscope (only)
Adapter, Pressure Test	C7695507	Periscope (only)
Fixture, Shock Test	F7680617	Components (only)
Adapter, Shock Test	F7695431	Components (only)

4.6 Test methods and procedures.

4.6.1 Storage temperatures.- The periscope shall be subjected to each extreme temperature in 3.3.1 until its exposed surface is thermally stabilized at each extreme temperature. Upon completion of both exposures, the periscope shall be allowed to thermally stabilize in an ambient temperature range of 60°F to 90°F. The periscope shall then be examined for evidence of physical damage and subjected to the following tests to determine conformance to 3.3.1.

4.6.2 Vibration.- Vibration equipment providing a holding medium and the amplitude, frequency and time duration of vibration 3.3.2 shall be utilized for this test. The periscope shall be vibrated in a vertical plane. Unless otherwise specified, the test equipment shall be in conformance with the "Test Facilities" requirements of MIL-F-13926 and 4.5.2.3.

4.6.2.1 Post vibration.- Subsequent to vibration and prior to conducting the remaining tests, the periscope shall be given a visual and tectile examination for physical defects resulting from poor workmanship or materials. In addition to the visual and tactile examination, the periscope shall be vigorously shaken by hand to detect any audible evidence of internal defects. Periscopes which exhibit no evidence of damage shall be subjected to the following tests to assure compliance with 3.3.2.

4.6.3 Cleanliness.- The optical surfaces of the periscope shall be examined for cleanliness in accordance with MIL-O-13830 utilizing the procedure herein to determine conformance to 3.4. Employing transmitted light, hold the periscope with the light source at the upper window and conduct the examination while viewing through the lower window. Repeat the procedure by illuminating the lower window while viewing through the upper window.

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4.6.4 Sealing.- The periscope shall be subjected to an external pressure as specified in 3.5. The test shall be accomplished by means of a special testing device with adapter, capable of producing the required pressure for the time specified. This device and method shall be comparable to that depicted on Drawings F7680673, "Fixture Pressure Testing", and adapters for the Pressure Test Fixture F7695490 and C7695507. The pressure drop and time period shall not exceed the requirements of 3.5.

4.6.5 Performance.- Tests for collimation, image tilt and spherical power and astigmatism shall be accomplished by means of special testing devices (see 4.5.2.3) and a calibrated reticle or wall target. If a wall target is employed, mil size shall be computed by the formula: MIL size = $\frac{\text{Target Distance (in inches)}}{1000}$. Test methods and procedures comparable to those depicted on Drawings D7560124, F7695464, and F7695508 shall be employed as applicable.

4.6.5.1 Collimation.- This test shall be conducted with the periscope mounted in the special testing device. Observe that the optical line of sight, through the periscope, is depressed 2° 15' (40 mils) with respect to a line of sight normal to the mounting surface. While viewing through the ocular of the auxiliary telescope, observe that its horizontal crosshairs bisect the horizontal arms of the target cross - within the horizontal tolerance marks for the specified 2° 15' depression, and also that the telescope's vertical crosshairs bisect the vertical target arms - within the specified tolerance - to determine conformance to 3.6.1 in azimuth.

4.6.5.2 Image tilt.- Utilize a test set-up similar to 4.6.5 and auxiliary telescope having a vertical reticle line. The field of view (FOV) of the auxiliary telescope shall be subtend an angle greater than 80 mils. Target markings shall indicate over an 80 mil FOV a displacement from its vertical line which subtends a 1.5 mil angle. The vertical reticle line and vertical target line shall now be made plumb. To determine conformance with 3.6.3, establish coincidence between the vertical reticle line of the auxiliary telescope and the vertical target line - while sighting through the periscope - at one extremity of the 80 mil FOV. Observe across the 80 mil FOV no displacement in target's vertical line image that will subtend an angle greater than 1.5 mil.

4.6.5.3 Spherical power and astigmatism.- Utilize a four line resolution chart as depicted in MIL-O-13830 and a milli-dioptometer (see 4.5.2.3) for this test. Employ the set-up and operating instructions on Drawing D7560124 as applicable to determine conformance to 3.6.2 in any two target line sets which are 90° with respect to each other.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The Periscope: M23 shall be packaged, packed and marked in accordance with Packaging Data Sheet 7633306. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The Periscope: M23 formerly designated "T38" is a one power periscope that locks on the forward assembly of the Telescope Mount: M100, formerly designated "T169E1" and over the objective assembly of the Elbow Telescope: M101. It is utilized to off set the line of sight of the elbow telescope to eliminate interference to the line of sight caused by the evacuator chamber at the muzzle of the weapon. It is intended for use in conjunction with Elbow Telescope: M101 for direct sighting in self propelled howitzers.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

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Project No. 1240-A047

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